Brunnermeier&Sannikov(2014) model derivation

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May, 8 2023

Note 1. *There are omitted predetermined setting and adding "*" means as the same in Brunnermeier and Sannikov(2014)*

Suppose we have known the state variables of q_t and K_t

$$\mathrm{d}k_t = (\Phi(\iota_t) - \delta)k_t\mathrm{d}t + \sigma k_t \mathrm{d}Z_t \tag{1}$$

$$\mathrm{d}q_t = \mu_t^q q_t \mathrm{d}t + \sigma_t^q q_t \mathrm{d}Z_t \tag{2}$$

Based on the Ito's Product rule of geometry Brownian Motion:

$$\frac{\mathrm{d}(k_t q_t)}{k_t q_t} = (\Phi(\iota_t) - \delta + \mu^q + \sigma\sigma_t^q)dt + (\sigma + \sigma_t^q)dZ_t \tag{3}$$

$$\frac{\mathrm{d}n_t}{n_t} = x_t \mathrm{d}r_t^k + (1 - x_t)r\mathrm{d}t - \frac{\mathrm{d}c_t}{n_t} \tag{4}$$

denote N_t is the sum of all experts, and

$$\frac{\mathrm{d}n_t}{n_t} = x_t \mathrm{d}r_t^k + (1 - x_t)r\mathrm{d}t - \frac{\mathrm{d}c_t}{n_t}$$

and $X_t = x_t = \psi(q_t K_t / N_t) = \frac{\psi}{\eta}$ postulate the state equation of N_t. Let $N_t = m \cdot n_t$

$$\begin{aligned} \mathrm{d}N_t = d(m \cdot n_t) &= mn_t x_t \mathrm{d}r_t^k + mn_t(1 - x_t)r\mathrm{d}t - m\mathrm{d}c_t \\ &= N_t x_t \mathrm{d}r_t^k + N_t(1 - x_t)r\mathrm{d}t - \mathrm{d}C_t \\ &= N_t \psi q_t K_t(\mathrm{d}r_t^k - r\mathrm{d}t) + N_t r\mathrm{d}t - \mathrm{d}C_t \end{aligned} \tag{5}$$

where $\frac{d(k_t q_t)}{k_t q_t}$ is the expert return on capital at time t. And $\frac{d(q_t K_t)}{q_t K_t}$ represents the income level of all asset holders at time t, where Expert and Household account for ψ,$1 - \psi$ respectively. We notice

that the total repay including the part of dividend.Expert's total return dr_t^k is

$$\begin{aligned} \mathrm{d}r_t^k &= \frac{a-\iota}{q_t}dt + \frac{d(k_t q_t)}{k_t q_t} \\ &= \frac{a-\iota}{q_t}\mathrm{d}t + (\Phi(\iota_t) - \delta + \mu^q + \sigma\sigma_t^q)\mathrm{d}t + (\sigma + \sigma_t^q)\mathrm{d}Z_t \end{aligned} \tag{6}$$

Similarly, we can get the total return rate $d\underline{r}_t^k$ belonging to household

$$\begin{aligned} \mathrm{d}\underline{r}_t^k &= \frac{\underline{a}-\underline{\iota}}{q_t}dt + \frac{\mathrm{d}(k_t q_t)}{k_t q_t} \\ &= \frac{\underline{a}-\underline{\iota}}{q_t}dt + (\Phi(\underline{\iota}_t) - \underline{\delta} + \mu^q + \sigma\sigma_t^q)\mathrm{d}t + (\sigma + \sigma_t^q)\mathrm{d}Z_t \end{aligned} \tag{7}$$

do the partial derivative of expert's and household's return((6) and (7)) about ι such that we can get the optimal condition as follows.

$$\begin{aligned} \max_{\iota} \Phi(\iota) - \frac{\iota}{q_t} \\ F.O.C. \Rightarrow \Phi'(\iota) = \frac{1}{q_t} \\ \iota_t = \underline{\iota_t} = \iota(q_t) \end{aligned}$$

Therefore, we can combine the optimal investment rate ι derived from (6) and (7). But the original parameters a, δ stay same. The physical asset held by household is

$$1 - \psi_t = \frac{1}{K_t}\int_{\mathbb{J}} \underline{k_t^j}\mathrm{d}j$$

Optimal Asset Selection

Under the equilibrium condition, the expectation return of household will less or equal to free-risk rate.

$$E_t[\mathrm{d}\underline{r}_t^k]/dt = \frac{\underline{a}-\iota}{q_t} + \Phi(\iota_t) - \underline{\delta} + \mu^q + \sigma\sigma_t^q \leq r, \quad 1 - \psi_t > 0 \tag{H}$$

under the meaning of equilibrium condition: if the return over r i.e risk-free rate then household will not lend to experts such that lift the investment rate which will cut off the return in the meanwhile.

As for expert's optimal decision function, we need to consider the sum of infinite expected utility.

$$\theta_t n_t \equiv E_t\left[\int_t^\infty e^{-\rho(s-t)}dc_s\right] \tag{10*}$$

where θ_t is the stochastic opportunities set. Furthermore, we consider such a process,

$$\Theta_t = \int_0^t e^{-\rho s} n_s d\zeta_s + e^{-\rho t}\theta_t n_t$$

employing the Integration by parts formula:

$$\mathrm{d}\Theta_t = e^{-\rho t}(n_t \mathrm{d}\zeta_t - \rho\theta_t n_t \mathrm{d}t + \mathrm{d}(\theta_t n_t)) \tag{8}$$

from $E(\mathrm{d}\Theta_t) = 0$ we could know Θ_t is a martingale. And the drift term of $d\Theta_t$ equals to zero based on the property of martingale. plug into (10*)

$$\rho\theta_t n_t = n_t d\zeta_t + E[d(\theta_t n_t)]$$

when n_t follows previous equation, also this economic system satisfies the transversality condition $E(e^{-\rho t}\theta_t n_t) \to 0$. Further, it can be proved that the Bellman equation is established by inverse proof, and the optimal strategy under constraints can be obtained:

$$\begin{aligned} \rho\theta_t n_t dt &= \max_{\hat{x_t}\geq 0, d\hat{\zeta}\geq 0} n_t d\hat{\zeta}_t + E[d(\theta_t n_t)] \\ s.t. \quad \frac{dn_t}{n_t} &= \hat{x}_t dr_t^k + (1-\hat{x}_t) r dt - d\hat{\zeta}_t \end{aligned} \tag{9}$$

Proposition 1. *Separate the above state variable processes and consider the stochastic process of θ.*

$$\frac{d\theta}{\theta} = \mu_t^\theta dt + \sigma_t^\theta dZ_t.$$

(i) $\theta \geq 1$, and only consume when $\theta = 1$
(ii) μ_t^θ is the drift term of θ at time t.

$$\mu_t^\theta = \rho - r \tag{E}$$

(iii) any $x_t > 0$

$$\frac{a-\iota(q_t)}{q_t} + \Phi(\iota(q_t)) - \delta + \mu^q + \sigma\sigma_t^q - r = -\sigma_t^\theta(\sigma + \sigma_t^q) \tag{EK}$$

when $x_t = 0$, $E[dr_t^k]/dt - r \leq -\sigma_t^\theta(\sigma + \sigma_t^q)$
(iv) The transversality $E[e^{-\rho t}\theta_t n_t] \to 0$

Proof. According to the hypothesis, Hamilton equation is constructed:

$$\begin{aligned} H &= n_t \mathrm{d}\hat{\zeta}_t + E[d(\theta_t n_t)] + \theta_t(n_t\mu_t^n) + \sigma_t^\theta\theta_t(n_t\sigma_t^n) \\ &= n_t \mathrm{d}\hat{\zeta}_t + e^{-\rho t}(n_t \mathrm{d}\hat{\zeta}_t - \rho\theta n_t) + \theta_t\left(-n_t \mathrm{d}\hat{\zeta}_t + n_t\hat{x}_t(\frac{a-\iota}{q_t} + \Phi(\iota_t) - \delta + \mu^q + \sigma\sigma_t^q) + n_t(1-\hat{x}_t)r\right) \\ &+ \sigma_t^\theta\theta_t(n_t\hat{x}_t(\sigma + \sigma_t^q)) \end{aligned}$$

where θ and σ^θ is the co-states and

$$E[\mathrm{d}(\theta_t n_t)] = e^{-\rho t}(n_t \mathrm{d}\hat{\zeta}_t - \rho\theta n_t)$$

$$dr_t^k = (\frac{a-\iota}{q_t} + \Phi(\iota_t) - \delta + \mu^q + \sigma\sigma_t^q)dt + (\sigma + \sigma_t^q)dZ_t \tag{10}$$

$$\frac{\mathrm{d}\theta_t}{\theta_t} = \mu_t^{\theta_t}\mathrm{d}t + \sigma^{\theta_t}\mathrm{d}Z_t = -r\mathrm{d}t - \sigma_t^\theta \mathrm{d}Z_t \tag{11}$$

where the first order conditions of $\iota_t, \mathrm{d}\hat{\zeta}_t, \hat{x}_t$ are

$$\Phi'(\iota_t) = \frac{1}{q_t} \tag{12}$$

$$e^{-\rho t} = \theta_t - 1 \tag{13}$$

$$\frac{a-\iota}{q_t} + \Phi(\iota_t) - \delta + \mu^q + \sigma\sigma_t^q - r = -\sigma_t^\theta(\sigma + \sigma_t^q). \tag{14}$$

And θ_t satisfies BSDE condition:

$$\mathrm{d}\theta_t = -\frac{\partial H}{\partial n_t}\mathrm{d}t - \sigma_t^\theta \theta_t \mathrm{d}Z_t$$

where drift term θ_t is

$$\begin{aligned}
\mu_t^\theta \theta_t &= \frac{\partial H}{\partial n_t} \\
&= -\theta_t \left[\hat{x}_t(\frac{a-\iota}{q_t} + \Phi(\iota_t) - \delta + \mu^q + \sigma\sigma_t^q) + (1-\hat{x}_t)r + \hat{x}_t\sigma_t^\theta(\sigma + \sigma_t^q)\right] + \mathrm{d}\hat{\zeta}_t + e^{-\rho t}(\mathrm{d}\hat{\zeta}_t - \rho\theta) \\
&= -r\theta_t + \rho\theta_t
\end{aligned}$$

Thus the law motion of θ_t is as follows

$$\frac{\mathrm{d}\theta_t}{\theta_t} = (\rho - r_t)\mathrm{d}t + \sigma^\theta \mathrm{d}Z_t$$

when $\hat{x}_t = 0$ the corresponding F.O.C. transfer to loose constraint

$$\frac{a-\iota}{q_t} + \Phi(\iota_t) - \delta + \mu^q + \sigma\sigma_t^q \leq -\sigma_t^\theta(\sigma + \sigma_t^q).$$

based on F.O.C. of $d\hat{\zeta}$ we can know that $\theta_t \geq 1$. When $\theta_t \to 1$, the computed $e^{-\rho t}$ converges to 0, which means that the value of future consumption discounted to the current period tends to 0, so the expert will choose the current consumption.

□

Wealth Distribution

(6) combines with (7) we can get the formula as follows according to Ito formula:

$$\begin{aligned}\frac{\mathrm{d}(q_tK_t)}{q_tK_t} &= \mathrm{d}r_t^k - \frac{a-\iota(q_t)}{q_t}\mathrm{d}t - (1-\psi)(\underline{\delta}-\delta)\mathrm{d}t \\ &= \frac{\mathrm{d}(k_tq_t)}{k_tq_t} - (1-\psi)(\underline{\delta}-\delta)\mathrm{d}t \\ &= (\Phi(\iota_t) - \delta + \mu^q + \sigma\sigma_t^q - (1-\psi)(\underline{\delta}-\delta))dt + (\sigma+\sigma_t^q)\mathrm{d}Z_t\end{aligned} \tag{15}$$

η_t denotes the ratio of expert's wealth to total asset, in math:

$$\eta_t \equiv \frac{N_t}{q_tK_t} \in [0,1]$$

based on (15) we can get the law motion of $d(\frac{1}{q_tK_t})$.

$$\frac{\mathrm{d}(\frac{1}{q_tK_t})}{\frac{1}{q_tK_t}} = -\mathrm{d}r_t^k + \frac{a-\iota(q_t)}{q_t}\mathrm{d}t + (1-\psi)(\underline{\delta}-\delta)\mathrm{d}t + (\sigma+\sigma_t^q)^2\mathrm{d}t \tag{16}$$

Combined with equation N_t i.e. (5) and (16). we can infer the state variable of η_t.

$$\begin{aligned}\mathrm{d}(\frac{1}{q_tK_t}) &= \frac{1}{q_tK_t}\left(-\mathrm{d}r_t^k + \frac{a-\iota(q_t)}{q_t}\mathrm{d}t + (1-\psi)(\underline{\delta}-\delta)\mathrm{d}t + (\sigma+\sigma_t^q)^2\mathrm{d}t\right) \\ \mathrm{d}N_t &= N_t\left(\psi q_tK_t(\mathrm{d}r_t^k - r\mathrm{d}t) + r\mathrm{d}t - \mathrm{d}\zeta_t\right) \\ \mathrm{d}\eta_t &= (\mathrm{d}N_t)\frac{1}{q_tK_t} + N_t\mathrm{d}(\frac{1}{q_tK_t}) + \psi q_tK_t(\sigma+\sigma_t^q)\frac{-1}{q_tK_t}(\sigma+\sigma_t^q)\mathrm{d}t \\ &= \eta_t\left(\psi(\frac{1}{\eta_t}-1)(\mathrm{d}r_t^k - r\mathrm{d}t) - \mathrm{d}\zeta_t + \frac{a-\iota(q_t)}{q_t}\mathrm{d}t + (1-\psi)(\underline{\delta}-\delta)\mathrm{d}t\right) + (\eta-\psi)(\sigma+\sigma_t^q)^2 \\ &= (\psi_t-\eta_t)(\mathrm{d}r_t^k - r\mathrm{d}t - (\sigma+\sigma_t^q)^2\mathrm{d}t) + \eta_t\left(\frac{a-\iota_(q_t)}{q_t} + (1-\psi_t)(\underline{\delta}-\delta)\right)\mathrm{d}t - \eta_t\mathrm{d}\zeta_t\end{aligned} \tag{14*}$$

in line with (EK) and dr_t^k we can infer:

$$\mathrm{d}r_t^k - (\sigma+\sigma_t^q)\mathrm{d}Z_t = -\sigma_t^\theta(\sigma+\sigma_t^q)\mathrm{d}t + r\mathrm{d}t$$

and transfer the above formula to

$$\mathrm{d}\eta_t/\eta_t = \frac{\psi_t-\eta_t}{\eta_t}\left(-(\sigma+\sigma_t^q)(\sigma_t^\theta+\sigma+\sigma_t^q) + \frac{a-\iota_(q_t)}{q_t} + (1-\psi_t)(\underline{\delta}-\delta)\right)\mathrm{d}t - \mathrm{d}\zeta_t + \frac{\psi_t-\eta_t}{\eta_t}(\sigma+\sigma_t^q)\mathrm{d}Z_t \tag{17}$$

Thus, the drift of $\mathrm{d}\eta_t$

$$\mu_t^\eta = -\sigma_t^\eta(\sigma+\sigma_t^q+\sigma_t^\theta) + \frac{a-\iota(q_t)}{q_t} + (1-\psi_t)(\underline{\delta}-\delta) \tag{18}$$

The volatility term is

$$\sigma_t^\eta = \frac{\psi_t - \eta_t}{\eta_t}(\sigma + \sigma_t^q) \tag{19}$$

Markov Equilibrium

Use Ito's Lemma to $q(\eta_t)$

$$\mathrm{d}q(\eta) = (q'(\eta)(u_t^\eta \eta) + \frac{1}{2}q''(\eta)(\sigma_t^\eta \eta)^2)\mathrm{d}t + \sigma_t^\eta q'(\eta)\eta dZ_t \tag{20}$$

corresponding term to the Brownian term of $\frac{\mathrm{d}q_t}{q_t} = \mu_t^q \mathrm{d}t + \sigma_t^q \mathrm{d}Z_t$

$$q(\eta)\sigma_t^q = \sigma_t^\eta q'(\eta)\eta \tag{21}$$

(19) plug in above equation we will get

$$\sigma_t^q q(\eta) = q'(\eta)(\psi - \eta)(\sigma + \sigma_t^q) \quad \Rightarrow \quad \sigma_t^q = \frac{q'(\eta)}{q(\eta)} \frac{(\psi - \eta)\sigma}{1 - \frac{q'(\eta)}{q(\eta)}(\psi - \eta)}. \tag{22}$$

In the meanwhile from (EK) and (H) we can infer (C2)

$$\frac{a - \iota(q_t)}{q_t} + \Phi(\iota(q_t)) - \delta + \mu^q + \sigma\sigma_t^q + \sigma_t^\theta(\sigma + \sigma_t^q) = r \tag{EK}$$

$$\frac{\underline{a} - \iota(q_t)}{q_t} + \Phi(\iota(q_t)) - \underline{\delta} + \mu^q + \sigma\sigma_t^q \leq r \tag{H}$$

$$\frac{a - \underline{a}}{q(\eta)} + \underline{\delta} - \delta + (\sigma + \sigma_t^q)\sigma_t^\theta \geq 0 \tag{C2}$$

when $\psi < 1$ or $1 - \psi > 0$ i.e. household holds physical asset as well. When $q(\eta), q'(\eta), \theta(\eta) > 0$ and $\theta'(\eta) < 0$, we can infer that $\sigma_t^q, \sigma_t^\eta > 0$ will go up with ψ and $\sigma_t^\theta < 0$ goes down with ψ. $q(\eta)$ and ψ are unrelated. Only consider the part of $(\sigma + \sigma_t^q)\sigma_t^\theta$. Thus the inequality (C2)'s LHS will go down with ψ.

parameter solutions

from the previously synthetic conditions we can infer that:

$$\begin{aligned}
\mu_t^\eta &= -\sigma_t^\eta(\sigma + \sigma_t^q + \sigma_t^\theta) + \frac{a - \iota(q(\eta))}{q(\eta)} + (1 - \psi)(\underline{\delta} - \delta), \\
\mu_t^q &= r - \frac{a - \iota(q(\eta))}{q(\eta)} - \Phi(q(\eta)) + \delta - \sigma\sigma_t^q - \sigma_t^\theta(\sigma + \sigma_t^q), \\
\mu_t^\theta &= \rho - r
\end{aligned} \tag{19*}$$

Similar with (20) we can derivative $\theta(\eta)$ employing Ito Lemma:

$$\mathrm{d}\theta(\eta) = \left(\mu_t^\eta \theta'(\eta) + \frac{1}{2}(\eta\sigma_t^\eta)^2\theta''(\eta)\right)\mathrm{d}t + \theta'(\eta)\sigma_t^\eta\eta\mathrm{d}Z_t \tag{23}$$

use the corresponding term to $\frac{\mathrm{d}\theta}{\theta} = \mu_t^\theta \mathrm{d}t + \sigma_t^\theta \mathrm{d}Z_t$'s Brownian Term

$$\sigma_t^\theta = \frac{\theta'(\eta)}{\theta(\eta)}\sigma_t^\eta\eta \tag{24}$$

And we need to consider the corresponding volatility term

$$\theta(\eta)\mu_t^\theta = (\eta\mu_t^\eta)\theta(\eta)' + \frac{1}{2}(\sigma_t^\eta\eta)^2\theta(\eta)'' \Rightarrow \theta(\eta_t)''(\eta) = 2(\theta(\eta)\mu_t^\theta - \theta(\eta)'\mu_t^\eta)/(\sigma_t^\eta\eta)^2 \tag{25}$$

$$q(\eta)\mu_t^q = (\eta\mu_t^\eta)q(\eta)' + \frac{1}{2}(\sigma_t^\eta\eta)^2 q(\eta)'' \Rightarrow q(\eta_t)''(\eta) = 2(q(\eta)\mu_t^q - q(\eta)'\mu_t^\eta)/(\sigma_t^\eta\eta)^2 \tag{26}$$

You can get it with a little transformation:

$$q''(\eta) = \frac{2[\mu_t^q q(\eta) - q'(\eta)\mu_t^\eta\eta]}{(\sigma_t^\eta)^2\eta^2} \quad and \quad \theta''(\eta) = \frac{2[\mu_t^\theta\theta(\eta) - \theta'(\eta)\mu_t^\eta\eta]}{(\sigma_t^\eta)^2\eta^2} \tag{19*}$$

Boundary condition

1. under $\eta = 0$ i.e. household hold all asset, which $q(0) = \underline{q}$ guarantees
2. proposition1 when $\theta = 1$ expert choose consuming. we denote η^* as the consumption point, thus $\theta(\eta^*) = 1$
3. $q'(\eta^*) = 0, \theta'(\eta^*) = 0$ means the economy is stable. If $q'(\eta^*) < 0$ will make the experts holding assets suffer loss. If $\theta'(\eta^*) < 0$ would violate proposition1.
4. $\theta(0) = \infty$

Combining the above conditions, we can solve the unknown boundary PDE problem. The specific method is as described in the article: the equilibrium domain is $[0, \eta^*]$, the function boundary is as following:

$$q(0) = \underline{q}, \quad \theta(\eta^*) = 1, \quad q'(\eta^*) = 0, \quad \theta'(\eta^*) = 0, \quad and \quad \lim_{\eta\to 0} = \infty \tag{27}$$

make (C2) satisfied, i.e. the return rate of household in (H) equals to free-risk rate. expert also choose consuming.

$$\frac{a - \underline{a}}{q(\eta)} + \underline{\delta} - \delta + (\sigma + \sigma_t^q)\sigma_t^\theta = 0, \tag{C2*}$$

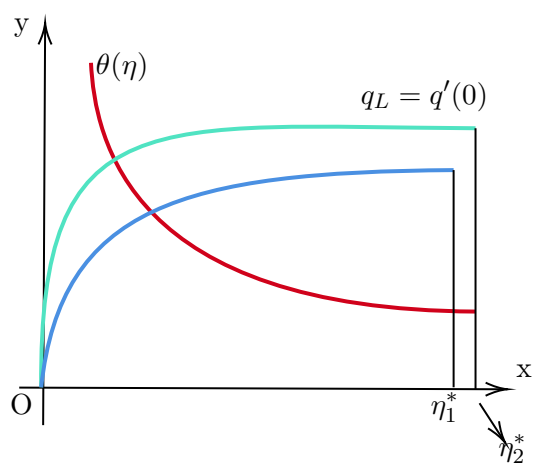
where

$$\sigma_t^\eta\eta = \frac{(\psi - \eta)\sigma}{1 - (\psi - \eta)q'(\eta)/q(\eta)}, \quad \sigma_t^q = \frac{q'(\eta)}{q(\eta)}\sigma_t^\eta\eta \quad and \quad \sigma_t^\theta = \frac{\theta'(\eta)}{\theta(\eta)}\sigma_t^\eta\eta \tag{12-14}$$

in the domain $\psi \in (\eta, \eta + q(\eta)/q'(\eta)$ find η which satisfies (C2*) if the found point out of boundary i.e. $\psi > 1$ which means the $\eta^{*'}$ in $\psi'(\eta_t^{*'})$ less than η^* in $\theta(\eta^*)$ and $q(\eta^*)$. Let $\psi = 1$ and calculate (12-14) to get $\theta(\eta)$ and $q(\eta)$ again. At last, calculate eq (19*)

Detailed algorithm

- because $\eta = 0$ is a singular point we need to decide the endogenous η matching the boundary condition of 0 and η^*.
- follow with (27) and set the initial condition $q(0) = \underline{q}, \quad \theta(0) = \theta(\eta^*) = 1$ and $\theta(0) = -10^{10}$
- assume $q_L = 0$ and $q_H = 10^{15}$
- Let $q'(0)$equal to $(q_L q_H)/2$as a condition to enter the solution of the ordinary differential equation to get $q(\eta)$and $\theta(\eta)$each calculation of the ordinary differential equation can be further calculated to get different η^*. See if any one of them meets its boundary conditions. If $q'(eta)$ to reach zero, shows that the set of initial conditions make falling too fast, let $q_L = q'(0)$if not $q'0(eta)$ arrive first, then make $q_H = q'(0)$.
- convergence after multiple times $(q_L = q_H)$
- at last need to make adjustment of $\theta(\eta^*)$ such that $\theta(\eta^*) = 1$.



solve ψ

Enter the loop and put the updated initial value condition into the function for iterative solution. in fnct, we need to get the value of Φ and ι from investment fnct. Set the left- right- boundary of ψ

to η and $\min(1, \eta + q(\eta)/q'(\eta))$ respectively. Iterating in the iterator, also let $\psi = (\psi_L \psi_R)/2$, and then solve under the given function.

$$amp = 1 - q'(0)(\psi - \eta)/q(0) \tag{28}$$

$$\sigma^\eta \eta = \sigma(\psi - \eta)/amp \tag{29}$$

$$\sigma^\theta = \sigma^\eta \eta \theta'(0)/\theta \tag{30}$$

$$rp = -\sigma^\theta(\sigma + \sigma^q) \tag{31}$$

$$hp = (\underline{a} - a)/q + \delta - \underline{\delta} + rp \tag{32}$$

If the risk premium of household is greater than 0, it indicates that household will hold more assets. It will eventually reach equilibrium as ψ changes. We end up with scalar ψ. Thus, you can put the solved ψ into $\sigma^\eta \eta$ and $\sigma^\eta \eta$ into σ^θ, σ^q. Further, expressions for μ^q, μ^η can be solved.

$$\mu^q = r - (a - \iota)/q(0) - \Phi + \delta - \sigma\sigma^q + rp \tag{33}$$

$$\mu^\eta \eta = -(\psi - \eta)(\sigma + \sigma^q(0))(\sigma + \sigma^q + \sigma^\theta) + \eta(a - \iota)/q(0) + \eta(1 - \psi)(\underline{\delta} - \delta) \tag{34}$$

$$q'' = 2((\mu^q q(0) - q'(0))\mu^\eta \eta)/\sigma^\eta \eta^2 \tag{35}$$

$$\theta'' = 2((\rho - r)\theta(0) - \theta'(0)) \tag{36}$$

Finally, put the parameters to be solved into $q''(\eta)$, you can solve the ordinary differential equation about $q(\eta)$, and then put the resulting $q(\eta)$ expression into θ'' to solve.